Acute Diarrhea

PAGE
DATE

* small intestinal diarrhea = small bowel (No deudenum or jejunum involved)
  . Large volume in acute / chronic

* Diarrhea means secretion > absorption
* 1-2L from exogenous sources

(GIT)

* Large intestinal = large bowel = colonic diarrhea (sigmoid, rectum, colon)
  . small volume in acute / chronic

① . Invasion (cut / disruption in mucosa) cause abdominal pain, fever where inflammatory cytokines are secreted
∴ fever, bloody diarrhea, pain meas organism is invasive

② . Produce toxin → Irritation
no invasion intact mucosa
irritation make mucosa unable to absorb
. Must have vomiting with diarrhea

PAGE
DATE

* Most of the causes of acute diarrhea are infectious contaminated food and water

* causes differ acc. to onset
* For example if caused early (4-6hr) after food → preformed toxin → Staph (most common)

* Voluminous diarrhea caused by (Imp) vibrio cholera (bacterial) in small intestine
. secrete toxin cause large vol. watery diarrhea → severe dehydration, hypotension, shock.
"Rice watery diarrhea"
∴ Voluminous diarrhea is Characteristic to Vibrio cholera

** Acute diarrhea is self limiting except in some conditions need further investigation.
* Viruses in immunocompetent patients (no chronic diseases) :-
from common to less common ↓
- Rotavirus
- Adenovirus
- Norwalk virus

** Cytomegalovirus and herpes Simplex in Immunocompromised patients. (V. Imp.)

PAGE
DATE

(cont.V)

* GIT affection of salmonella ?? (Imp)
- Typhoid fever / Gastroenteritis Acute Diarrhea.
By contaminated food (food handlers)
Salmonella carrier (salmonella present in gall bladder) poor hand washing can spread it.
ttt :- ciprofloxacin injection, Quinolones
ياخد على حسب الحالة
duration, conc., dose بال
For example
Flagyl 250 → Giardia
750 → Amoeba
** Dysentry = blood in stool + burning sensation + sense of incomplete evacuation
. Organisim → ① Shigella dysentry
cause Basillary dysentery
(Acute diarrhea bloody, fever, toxic)
More common in Rural, developind countries.
ttt :- anti-bacterial

. ② Amoebic dysentry by Amoeba histolytica (parasite).
ttt :- flagyl
same manifestations almost but differ in ttt. عشان كده لازم افرق ما بينهم

Basillary (V.Imp)	Amoebic
* Bacterial cause shigella	* Parasitic cause Entamoeba histolytica
* Bloody diarrhea	* Bloody diarrhea
* High grade fever	* Low grade fever
* More motion 10-12 times	* less motion 8 times
* Antibiotic	* Anti parasitic Flagyl

* Gastroentritis by salmonella if not treated may enter in typhoid fever

* Yersinia enterolitica infect terminal ileum, cecum resulting in Abdominal pain, fever, Acute diarrhea so can be Misdiagnosed with → Acute Apendicitis (have more constipation rather than diarrhea) (Also have psoas sign not present in yersinia)

PAGE
DATE

(V.Imp)

* Staph aureus, Bacillus cerus are most common causes of food poisining present in canned food, processed meat

First vomiting (4hrs) then → Acute diarrhea

* Clostridium difficile ⟶ Pseudomembranous colitis

patient developed diarrhea with history of antibiotic esp pencillin, cephalosporin

C. defficile ⟵ اسمك Dgbtk colitis (large bowel affection) not enteritis (small intestine)

• presentation :- bloody diarrhea sometimes misdiagnosed with inflammatory bowel disease

بس الفرق ان C. difficile دي حاجة acute

Ht :- Vancomycin (drug of choice) / metronidazole (Flagyl)

vax

* C. perferinges → Acute diarrhea
ttt : مش عارف

** Cryptosporidium in immunocompromised
Any bacteria can affect immuno " .

* Parasitic causes
Giardia (no invasion) less severe than entamoeba.
Shigella, entamoeba invasive
- Giardia cause irritation of mucosa
→ watery diarrhea, most common symptom steatorrhea oilspots in stool, abdominal pain, Flatulance.
- Diagnosed by presence of giardia cyst in stool analysis.
- ttt : Flagyl 250mg 3times for 7days (self limiting)

DATE

* Entamoeba histolytica
infective agent (trophozoite, cyst)
Flagyl لما اجى
↓
clinically يخف
بس لسة في cyst في ال intestine تعمل
و ترجع تاني autoinfection
عشان كده مش بعالج بال Flagyl لوحده
tt: Flagyl + Diloxanide furate
بعالج بحاجتين عشان في
. invasive
. has 2 stages
. One attack cyst other attack trophozoite

Medication causes
no fever, no toxicity manifestation, history of taking certain drug
عشان كده اي عيان عنده diarrhea مهم اسأل:
. history of contaminated food
. " taking certain drug
. " travelling abroad
* Antibiotics (most common)
Augmentin, clavulinic acid, Zithromax

PAGE
DATE

* Antihypertensives
hydralazine, Ca channel blockers
shift ممكن اعمل

* New psychatric drugs
Carbamazepine

** Organic condition
يعني في disease حاصل جوا الجسم مأثر على الـ blood supply, capacity
abdominal pain, anemia, fever عنده يبقى
Nocturnal, diurnal complaints

** Functional condition
كل الأعراض سليمة بس الجهاز بيشتكي

Brain Gut Axis
serotonin secreted from brain affect mobility of intestine
its secretion, metabolism is affected by patient's mood (stress, anxiety, ...)
↓
Irritable Bowel Syndrome
(Functional Gut Disorde)
tt: Symptomatic treatment
- severity of symptoms mainly in the morning (Diurnal)

PAGE
DATE

. symptoms alternating
organic causes ll exclusion بعد ما نعمل
(IBS) اشخص

** IBD (Inflammatory Bowel disease)
↓
.Ulcerative colitis
.Chron's disease

* hematochezia : fresh blood from rectum

* Tachycardia, orthostasis more in cholera

* presence of blood → 4i
- invasive organism or
- inflammation or
- infiltration (neoplasm) or
- ischemia

* Acute diarrhea → Managment (Rehydration)
Chronic " → Evaluation (find a cause)

* Occult is identified by Occult blood test

PAGE
DATE

* ttt of entamoeba histolytica :-
Metronidazole 750mg 3times a day for 10 days
+
Diloxanide furate 3 times daily for 10 days

* Most bacterial patogens causing gastroenbritis
(Shigella, salmonella, Campylobacter, E-coli, yarsinia)
First drug of choice → Quinolones

** Some times short colonscopy may be needed.
Contraindicated in patients withe severe tendrness and abdominal pain
↓ لين ده عليه فية
Intraabdominal inflammation
↓
must not be done till acute condition is resolved.

** Drug induced disease occur in weeks of adminstration.

Example:-

PAGE
DATE

Woman took 4vials of Methotrexate after 1 week she developed severe bloodydiarrhea, stomatitis, colitis, anemia, thrombocytopenia, leucopenia from the drug.

ttt:- Antidote of Methotrexate (Folinic acid)

∴ Drug history is life saving.

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Clues for certain infections:

- Vomiting that begins within 6 hours of ingesting a food should suggest food poisoning caused by preformed toxin from bacteria such as S. aureus or B. cereus.
- If diarrhea disease begins within 8 to 14 hours of food ingestion, C. perfringens should be suspected. bacterial اي حاجه
- When the incubation period is longer than 14 hours and vomiting is also a significant symptom, along with the diarrhea, viral agents should be considered.
- Parasites that do not invade the intestinal mucosa, such as Giardia lamblia and Cryptosporidium, usually cause only mild abdominal discomfort. Giardiasis may be associated with mild steatorrhea, gaseousness, and bloating. (sense of fulness)
- Infection with invasive bacteria such as Campylobacter, Salmonella, and Shigella spp., and organisms that produce cytotoxins, such as C. difficile and enterohaemorrhagic E.

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Indications to investigate a case of Acute Diarrhea:

- Evaluation is indicated if symptoms are severe or prolonged, the patient appears "toxic," there is evidence of colitis (occult or gross blood in the stools, severe abdominal pain or tenderness, and fever), or empirical therapy has failed.
  ← esp with shigella, entamoeba
- Passage of many small-volume stools containing blood and mucus, temperature higher than 38.5° C, passage of more than six unformed stools in 24 hours, or a duration of illness longer than 48 hours, diarrhea with severe abdominal pain in a patient older than 50 years, diarrhea in older adults (older than 70 years)., young
  ← • CBC for severe anemia • stool analysis
- ** The physical examination Vital signs (including temperature and orthostatic evaluation of pulse and blood pressure) and signs of volume depletion (including dry mucous membranes, decreased skin turgor, and confusion) should be carefully evaluated.
  اول حاجة تتعمل
- A careful abdominal examination to evaluate tenderness and distention.

Investigation of a case of Acute Diarrhea:

- Stool analysis: Fecal leukocyte (bacterial) determination, Stool culture (bacterial mainly) for enteric pathogens. Stool examination for ova and parasites (parasitic). Stool evaluation for fecal leukocytes (or lactoferrin, a by-product of white blood cells) is a useful initial test, because it may support a diagnosis of inflammatory diarrhea. If the test is negative, stool culture may not be necessary, but culture is indicated if the test is positive.

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